PIED-PIPING IN THE EXTERNAL MERGE IN COLLINS CONJUNCTIONS*

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Starting from an analysis of *Sue invited John and maybe Mary*, this paper shows that when the category of X should be merged with Y, X's containing clause seems to be pied-piped. The paper argues that X has covertly moved out of the local clause. The analysis applies to some other interrupting clauses, including Transparent Free Relatives.

1. Introduction

There are different types of parentheticals, and not all of them need an adjacent prosodic break (Dehé 2014). One type that is consistently lack of such a break is seen in a Collins Conjunction (CC), such as (1) (Collins 1988). In a CC, a nominal conjunct contains an adverb. I call such a nominal conjunct (the italic part) Adv Conjunct. Similar examples in Spanish and Dutch are seen in Vicente (2013) and Zwart (2023), respectively.

(1) Sue invited John and *maybe Mary*.

The weak reading of (1) is that John was invited, and maybe Mary was also invited. The involvement of the individual denoted by the Adv Conjunct in the event is not entailed. The strong reading of (1) is that John and another person were invited, and maybe the other person was Mary. The involvement of an individual denoted by the Adv Conjunct in the event is entailed, although her identity is not clear to the speaker.

I answer two questions. A: how a sentential adverb is combined with a nominal in an Adv Conjunct; B: how the different readings of a CC are computed in syntax. I will argue that an Adv Conjunct is a reduced clause, and thus the adverb modifies the clause; also, this clause occurs in a nominal position, because the contained nominal covertly adjoins to a categoryless expression. Moreover, different readings of a CC correlate with the occurrence of a silent BE or HAVE in the reduced clause.

My analysis of CCs does not support the hypothesis that nominal coordination must be derived from clausal coordination (cf. Chomsky 1957). This research also exhibits the explanatory power of a categoryless functional element (Zhang 2003).

*I am grateful to Chris Collins, Jim Wood, James Huang, Carlo Geraci, Elizabeth Cowper, Jila Ghomeshi, Iris Hsiao-Hung Wu, Sam Hsuan-Hsiang Wang, Jenny Trang Ho, other (anonymous) reviewers, and the audiences of the Yushan Colloquium and the CLA 2024 for their comments at various stages of the research. But none of them is responsible for the claims made here. All remaining errors are mine. This research has been partially supported by a grant from the National Science Council of Taiwan.

Actes du congrès annuel de l'Association canadienne de linguistique 2024.
Proceedings of the 2024 annual conference of the Canadian Linguistic Association.

In §2, I analyze the internal structure of Adv Conjuncts. In §3, I discuss the categorial mismatch between an Adv Conjunct and its position. In §4, I introduce other similar constructions. In §5, I explain certain general syntactic properties of the constructions. In §6, I challenge available alternative analyses. §7 concludes the paper.

2. The syntax of Adv Conjuncts

2.1 The existence of a reduced clause in a CC

Collins (1988: 19) notes that only a sentential adverb may occur in a CC, excluding other kinds of adverbs, e.g., manner adverbs (*secretly, carefully, reluctantly*), as shown in (2).

(2) *John and *clumsily Mary* avoided the teacher.

Syntactically, a sentential adverb cannot merge with a definite nominal. Semantically, such an adverb requires a propositional (type *t*) argument, i.e., a clause.

Moreover, the sentential adverb in an Adv Conjunct does not scope beyond the conjunct (Zwart 2023). We can see this in the dialogue in (3). B1 gives a negative response to the weak reading of the CC uttered by speaker A, i.e., whether Mary also went to the store, and B2 gives a negative response to the strong reading of the CC, i.e., whether another person that also went to store was Mary. But B3 is not felicitous. Obviously, the uncertainty expressed by A is the inclusion of Mary in the event, or the identity of the second person, rather than the occurrence of the event of going to the store. The acceptability contrast between B3 and the other two responses shows that the scope of the adverb in an Adv Conjunct cannot be external to the nominal coordination.

(3) A: John and maybe Mary went to the store. I'm not sure.
B1: But I'm sure: It's John alone who went to the store.
B2: But I'm sure: It is not Mary who went to the store with John. The other person was Sue.
B3: #But I'm sure: John and Mary indeed went to the store. I know they did not stay at home.

Thus, as claimed by Collins (1988) and Zwart (2023), the adverb in a CC must be sentential and nominal coordination-internal. Then, the adverb must modify a reduced clause. Accordingly, there must be two clauses in a CC construction: a Host Clause and a reduced clause, which I call Interrupting Clause (I-Clause, a term borrowed from Kluck 2011). (4a)'s Host Clause is (4b), and its I-Clause is (4c).

(4) a. John and perhaps Mary went to the store.
b. $[^{\text{Host Clause}}$ John and $_{\text{DP}}$_ went to the store]
c. $[^{\text{I-Clause}}$ perhaps … Mary …]

The rest of this section specifies that the I-Clause in a CC has a null BE or HAVE.

2.2 An individual *pro* in a BE clause

The I-Clause in a CC can be a reduced copular clause. Two major types of copular clauses are considered: predicational and specificational (Higgins 1979), as shown in (5a) and (5b), respectively. "The intuition about predicational clauses is that they predicate a property of the subject referent" (Mikkelsen 2011: 1807). (5a) tells us the novelist property of Jane Austen, but "The term specificational derives from the intuition that these clauses are used to specify who (or what) someone (or something) is, rather than to say anything about that person (or entity)." (Mikkelsen 2011: 1809) (5b) tells us who wrote a certain novel, not anything about that person (cf. Moro 1997, den Dikken 2006).

(5) a. Jane Austen was a novelist. [predicational clause]
b. The author of *Pride and Prejudice* was Jane Austen. [specificational clause]

In an NP CC such as (6a), the NP in the Adv Conjunct is predicative. Plausibly, the I-Clause is a predicational copular clause. If we spell out the syntactic form of the Adv Conjunct in (6a), it should be (6b). It has a null BE and a *pro* subject. The antecedent of this *pro* is the subject of the Host Clause.

(6) a. My [friend and *perhaps future colleague*] will also attend.
b. $[^{\text{Adv Conjunct}}$ perhaps [*pro* BE future colleague]]

As for a specificational clause, the post-copula nominal is referential. Similarly, in the strong reading of a CC like that in (7a), the nominal in the Adv Conjunct is also referential. (7a) means John, Bill and another person went to the store, and maybe that person was Mary. The I-Clause specifies who the third person is. I propose that the I-Clause in such a CC is a specificational clause. Then, the definite nominal follows a *pro* and a hidden copula. The *pro* is an anaphoric definite element, associated with a presupposed individual in the context. If we spell out the syntactic form of the strong Adv Conjunct in (7a), it should be (7b) (cf. Heringa 2011).

(7) a. John, Bill and *maybe Mary* went to the store.
b. $[^{\text{Adv Conjunct}}$ maybe [*pro* BE Mary]] (for the strong reading)

In summary, an NP CC has a reduced predicational copular clause, and a DP CC in its strong reading has a reduced specificational copular clause.

2.3 An eventive *pro* in a HAVE clause

I propose that the I-Clause in a CC can also be about the involvement of an individual in an eventuality. This is the case when a DP CC has a weak reading. In this reading, (8a) means that John and Bill went to the store, and this event may also have Mary involved. If we spell out the syntactic form of the Adv Conjunct in (8a), it should be (8b), which has a null HAVE and an eventive *pro* (marked as pro_v). This pro_v is an anaphoric definite element, taking the event denoted by the Host Clause as its antecedent.

(8) a. John, Bill and *maybe Mary* went to the store.
b. [$^{\text{Adv Conjunct}}$ maybe [pro_v HAVE Mary]] (for the weak reading)

In English, if the subject of a clause is eventive and the verb is the overt *have*, the clause needs the PP *in it*, for no clear syntactic reason, as seen in (9).

(9) The competition had Mary *(in it).

No one denies that an event can be encoded by an overt or null pronoun, as an argument in a clause (see Mittwoch 2005: 70 and Ramchand 2007: 478, among others).

In summary, an Adv Adjunct has a reduced BE clause with an individual *pro*, or a reduced HAVE clause with an eventive *pro*. It is such a clause that the adverb modifies.

3. The categorial mismatch of an I-Clause and its position

In this section, I specify the root category of an Adv Conjunct (§3.1), and then show a categorial mismatch of the Adv Conjunct and its surface position (§3.2).

3.1 The I-Clause is not contained in a nominal

How can the I-Clause in a CC fit in the position of a nominal conjunct? If a clause is contained in a nominal, it can be either a clausal complement or modifier of a nominal. But neither is true of the I-Clause in a CC.

There is no evidence for the I-Clause in a CC to be the complement of any noun or nominal head. We also rule out the possibility for the I-Clause to be a modifying clause, regardless of what kind of a modifying clause it is. First, the I-Clause cannot be a non-restrictive relative clause. If an expression has a non-restrictive clause, it entails the expression that does not have the non-restrictive relative clause. Thus, (10a) entails (10b). However, the CC example in (11a) does not entail the non-CC example in (11b).

(10) a. John and Mary, who perhaps was on her vacation, went to the store yesterday.
b. John and Mary went to the store yesterday.

(11) a. John and perhaps Mary went to the store yesterday.
b. John and Mary went to the store yesterday.

Second, the I-Clause in a CC is not a reduced restrictive relative clause, either, since it allows speaker-oriented adverbs (Collins 1988: 18), but a restrictive relative does not.

(12) a. *I know a woman [whose brother *frankly* is quite handsome].
b. The old man left all his money to John, Bill and *unfortunately Sue*.

Moreover, a relative clause can be extraposed to the right edge of the construction, as in (13a), but this is not possible for the I-Clause in a CC, as seen in (13b).

(13) a. Many paintings of young artists _ are on sale [$_{CP}$ which I like]
b. *John and _ went to the store, maybe Mary.

Third, the I-Clause in a CC starts with an adverb, it thus cannot be a free relative, which is headless and must start with a WH-form in English.

In summary, the root category of an Adv Conjunct is just a clause, not a nominal.

3.2 The apparent pied-piping of the I-Clause

If the I-Clause in a CC has no way to be embedded in a nominal, we need to explain how the clause appears in a nominal position. van Riemsdijk (1998, 2017) uses a Graft theory to derive a Transparent Free Relative (TFR, Wilder 1998) construction, such as (14), where the WH-clause occurs in a DP position.

(14) They put <u>what appeared to be a steak to me</u> on my plate.

It seems that the I-Clause of a CC is also grafted into a nominal position in the Host Clause. This Graft operation looks like pied-piping. Pied-piping is a phenomenon that when one element is probed in an operation, a larger element undergoes the operation. For example, when a WH element is internally merged with an element in the C-domain, it can carry its containing nominal or PP (Ross 1967). In (15), *which house* is probed by C, but it is the containing PP *in which house* that is moved.

(15) In which house does John live?

The internal merge of the PP in (15) seems to be parallel to the merging of an Adv Conjunct in a CC: in both cases, something extra is carried with the probed element. In (15), a quantifier is probed, but the containing PP is internally merged. In a CC, a nominal is probed, but the containing I-Clause is externally merged.

The special integration of the I-Clause into the Host Clause in CCs is exhibited in two aspects: the fully integration of the nominal in the I-Clause into the Host Clause and the non-integration of the rest part of the I-Clause.

The fully integration of the nominal in the I-Clause into the Host Clause can be seen in two points. First, the nominal exhibits all properties of a normal nominal conjunct. For example, if there are two singular conjuncts, a CC also triggers a plural agreement.

(16) John and *maybe Mary* {want/*wants} to go to the store.

Also, in (17), the coordinate nominal occurs in an accusative position, and nominal in the Adv Conjunct is also accusative.

(17) Having him injury free is a must for [Arsenal *and perhaps him*] not being too involved with the Brazilian Olympic team.
(https://paininthearsenal.com/2021/08/07/gabi-martinelli-ace-arsenal-gold-medal/)

Thus, the nominal contained in the I-Clause of a CC is categorially integrated into the Host Clause in syntax, exhibiting the syntactic properties of a nominal conjunct.

On the other hand, the other parts of the I-Clause are not integrated into the Host Clause in syntax. First, an I-Clause occurs in a non-clausal position. In (18a), for example, although the predicate *went to the store* rejects any (coordinate) clausal subject, as seen in (18b), the occurrence of the two ICs in the subject position is well-formed.

(18) a. Perhaps Mary and certainly John went to the store.
b. *[$_{CP}$ (Perhaps) Mary sang and (certainly) John danced] went to the store.

Second, the adverb in a CC does not interact with any element of the Host Clause (§2.1).

I conclude that other than the nominal that is probed by an element of the Host Clause, the rest part of the I-Clause in a CC is opaque to the Host Clause (also see §5.2).

Summarizing, in this section, I have argued that the root category of an Adv Conjunct is still a clause, not a nominal. Then, when a nominal is probed for by an element of the Host Clause, its containing clause seems to be pied-piped. The nominal in an I-Clause categorially matches the environment of the Host Clause, but it is interpreted in the I-Clause, and the whole I-Clause surfaces at a nominal position.

4. Beyond adverbs and beyond conjuncts

The category mismatch of a CC can be seen in many more constructions, including TFR constructions. In the following, I list various syntactic types of elements that are probed by an element of the Host Clause (such examples can be found in Lakoff 1974, Ernst 1984: 112, Kluck 2011, Bogal-Allbritten 2014, Griffiths 2015, Grosu 2023).

(19) a. Mary is meeting with *probably* a nurse practitioner.
Host Clause: [Mary is meeting with _$_{DP}$]
I-Clause: [Probably *pro* BE <u>a nurse practitioner</u>]

b. John is going to *I think it's* Chicago on Sunday.
Host Clause: [John is going to _$_{DP}$ on Sunday]
I-Clause: [I think it's <u>Chicago</u>]

(20) John and his *I hope* wife are coming to the party together.
Host Clause: [John and his _$_{NP}$ are coming to the party together]
I-Clause: [I hope *pro* BE <u>wife</u>]

(21) Bill Gates has a fortune of *I heard the news that CNN claims that it is* more than 100 billion dollars.
Host Clause: [Bill Gates has a fortune of _$_{\text{NumP}}$]
I-Clause: [I heard the news that CNN claims that it is more than 100 billion dollars]

(22) a. Bea was *you can imagine how* angry at Bob.
Host Clause: [Bea was _ $_{\text{predicative AP}}$ at Bob]
I-Clause: [you can imagine how angry *pro* BE]

b. Bea got herself a *you'll never guess how* expensive car.
Host Clause: [Bea got herself a _ $_{\text{attributive AP}}$ car]
I-Clause: [you'll never guess how expensive *pro* BE]

(23) He came out the next day, but I didn't get a chance to talk to him *what one might call* {privately/in a private way}.
Host Clause: [He came out the next day, but I didn't get a chance to talk to him _$_{\text{AdvP/PP}}$]
I-Clause: [what one might call {privately/in a private way}]

(24) a. Einstein's theory of special relativity – *I think* – was presented in his 1905 paper.
Host Clause: [Einstein's theory of special relativity _ $_{\text{VP}}$]
I-Clause: [I think [*pro* was presented in his 1905 paper]]

b. He felt my mother was *what he called* poisoning my mind.
Host Clause: [He felt my mother was _$_{\text{VP}}$]
I-Clause: [what he called poisoning my mind]

Therefore, CC is just one type of the constructions where an I-Clause appears in a non-clausal (DP, NP, AP, PP, AdvP, VP) position. I call such constructions Graft constructions. They all have a gap of a certain category in the Host Clause and the I-Clause contains and restricts the reading of an element of the category. Note that in (19), the argument nominal gap is satisfied by a predicative nominal in the I-Clause.

Then, a TFR is an I-Clause of a Graft construction where the probed element is a predicate in the I-Clause. As expected, the probed element occurs at the right-edge of the I-Clause (cf. Wilder 1998). Like an Adv Conjunct in a CC, a TFR is not next to an obligatory prosodic break, although it is a parenthetical (cf. Wilder 1998: 195).

An I-Clause is a secondary illocutionary expression to the Host Clause. It must be semantically related to the latter. (25) is not right, since the I-Clause is not semantically related to the Host Clause. In this sense, an I-Clause is similar to a secondary predicate, which also must be semantically related to (an element of) the main predication.

(25) *I drank [Bob bought a beer].

In summary, I-Clauses may fill the gaps of various categories in the Host Clauses.

5. Accounting for certain properties of the Graft Constructions

5.1 No direct merge of a Host Clause element and an I-Clause element

In a Graft construction, the I-Clause provides a semantic restriction on the contained element whose categorial feature is probed by the Host Clause. There is no direct merge between the probed element and the probing element in the Host Clause.

The lack of such a direct merge is seen in the fact that no idiom is formed across the clauses. If the probed expression is base-generated in the I-Clause, it may not form an idiomatic reading with any element out of the I-Clause, i.e., in the Host Clause. What the Host Clause probes for is just the formal features of a missing element, and thus there is no special lexical meaning to be established between the elements in the two clauses. In (26a), the verb *kicked* in the Host Clause is never directly merged with *the bucket* in the I-Clause, and thus the string *kicked the bucket* is never formed in any step of the derivation. More such examples are seen in (Grosu 2023: 16).

(26) a. #Bob kicked I think it was the bucket. (Kluck 2011: 106)
b. #Mary kicked {what must have been/what John thinks is} the bucket.
c. *It is raining cats and perhaps dogs.

The lack of a direct merge of a Host Clause element and the probed element in the I-Clause is also seen in the construction where the probed element is in the scope of a modality operator in the I-Clause. The Graft construction in (27a) (a CC construction) does not entail (27b), where no I-Clause occurs (also see (11)).

(27) a. John and *perhaps Sue* went to the store. $\nvDash$
b. John and Sue went to the store.

In (27b), *Sue* and *and* are merged directly, but in (27a), these two words do not merge directly. The form in (27b) is not a step to derive (27a). The fact that the TFR construction in (28a) (Grosu 2023: 2) does not entail (28b) is explained in the same way.

(28) a. This house is [what Bill would consider beautiful]. $\nvDash$
b. This house is beautiful.

In summary, the probing element in a Host Clause is not merged with the categorially probed element in the I-Clause directly.

5.2 The visibility of the I-Clause boundaries to the Host Clause

In the I-Clause in a Graft construction, other than the probed element by the Host Clause, no element is visible to the Host Clause syntactically. We have already seen two facts:

the I-Clause's clause category is ignored by the Host Clause (§3 and §4) and the adverb in the I-Clause is opaque to the Host Clause (§2.1). The following facts show another contrast regarding the I-Clause boundaries to the Host Clause: only the probed DP can be bound by a quantifier in the Host Clause, as in (29), and can show the Condition A, Condition B, and Condition C effect, as in (30), (31) and (32), respectively (Kluck 2011).

(29) a. No professor$_i$ believed the gossip about [I think it was <u>his$_i$ mistress</u>].
b. *No professor$_i$ taught, [{he$_i$/his$_i$ students} claimed it was <u>a boring class</u>].

(30) a. She$_i$ was [what can only be interpreted as <u>proud of herself$_i$</u>].
b. *She$_i$ was [what the stories about herself$_i$ suggested to be <u>rather arrogant</u>].

(31) a. *The professor$_i$ cites [I think it was <u>him$_i$</u>] primarily.
b. The professor$_i$ told Bea [he$_i$ didn't even remember himself how many boring <u>stories</u>].

(32) a. *He$_i$ cites [I think it was <u>the professor$_i$</u>] primarily.
b. He$_i$ had been kissing, [the professor$_i$ finally admitted it was <u>Bea</u>].

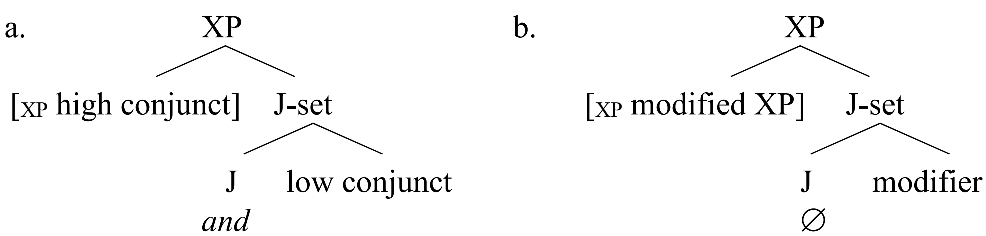
So far, no plausible analysis is available to explain the above contrasts between the probed and other elements in the I-Clause of a Graft construction.

5.3 The projection of the probed element

How is the visibility variation of the clausal boundaries of an I-Clause to the Host Clause represented in syntax? The hypothesis of the Graft operation or pied-piping in external merge is too vague to explain the derivation of a Graft construction. Instead, I use a J-theory to propose a derivation of the construction.

In Zhang's (2023, 2024) J-theory, a coordinator and a modification marker realize the categoryless functional element J (Junct). J takes a low conjunct or a modifier as its complement. The combination of J with its complement is a J-set, which is also categoryless, since the category of a complement does not project. The categoryless J-set is like a root. In a coordinate construction, it is the higher conjunct that projects and decides the category of the whole coordinate complex, as shown in (33a); and in a modification construction, it is the modified element that does the job, as shown in (33b).

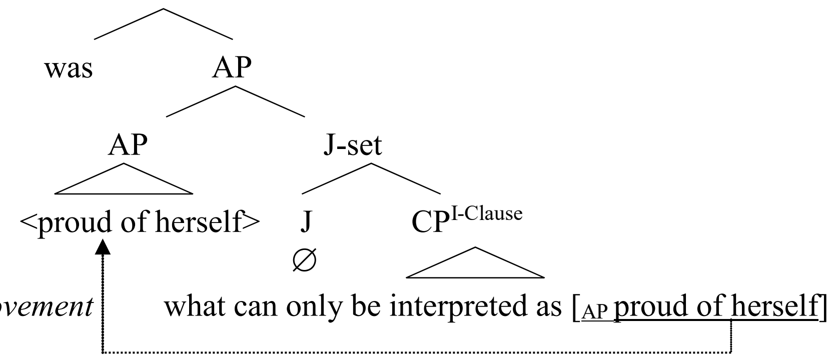
(33) a. b.



I propose that an I-Clause is merged with the functional element J, forming a categoryless J-set; and then the probed XP in the I-Clause moves covertly to merge with the J-set, and projects. For example, the structure of the predicate of (34a) is (34b).

(34) a. She$_i$ was [what can only be interpreted as proud of herself$_i$].

b.



In (34b), the AP [*proud of herself*] is probed by the Host Clause. Its high copy is out of the I-Clause, and thus local to the Host Clause. This explains the facts in §5.2 (note that in (32b), the matrix subject still c-commands the subject in the I-Clause; I leave the lack of the Condition C effect to future research). Moreover, the AP is base-generated in the I-Clause, but its high copy satisfies the selection of the probing element in the Host Clause. The word *was* in the Host Clause is merged with the AP that contains the I-Clause, but not with the probed AP *proud of herself* directly. This explains the facts in §5.1.

In this analysis, an element projects at its landing site only because it adjoins to a categoryless expression (J-set) (cf. Cecchetto & Donati 2015).

In this section, I have shown that in a Graft construction, the probing and the probed element do not merge together directly, and the probed element but not other elements in the I-Clause, is visible to the Host Clause. I have proposed a possible syntactic derivation of the construction. The main ingredients of the proposal include the occurrence of the categoryless functional element J, and the covert movement of the probed element from the I-Clause, to adjoining to the categoryless J-set. The projection of the high copy of the probed element explains the categorial mismatch of an I-Clause in its position.

6. Against other alternative analyses

I have challenged the alternative syntactic analyses of CCs: an I-Clause is embedded under a null nominal head as either a complement or a modifier (§3.1). I now show the inadequacies of other available analyses of various kinds of the Graft constructions.

6.1 The conjunction modification analysis

Collins (1988) focuses on the weak reading of a DP CC, claiming that the epistemic adverb modifies the conjunction *and*. One might assume that a conjunction denotes an inclusive relation, and if the speaker is not certain about whether the individual denoted by a nominal conjunct is included in the eventuality, such an epistemic adverb is used. But the conjunction modification assumption can be challenged.

First, an Adv Conjunct can be the first conjunct of a CC, as seen in (35). Collins recognizes that a coordinator and the conjunct to its right form a constituent in English. If so, the adverb *perhaps* in the first conjunct has no coordinator to modify in (35).

(35) Perhaps John and maybe Mary went to the store.

Second, the combination of an epistemic adverb and a nominal may occur in a non-conjunct position, and the combination expresses a similar meaning as an Adv Conjunct (see (19a)). In such a case, the adverb also has no coordinator to modify. I thus conclude that the adverb in an Adv Conjunct is not associated with the coordinator. Instead, as I proposed in §2, the adverb in a CC is associated with a reduced HAVE or BE clause.

6.2 The relative clause analysis

I have argued against restrictive and non-restrictive relative clause analyses of CCs in §3.1. Other kinds of Graft constructions are also not derived from any relative clauses. Tsubomoto & Whitman (2000) propose that (36) is from a relative clause construction.

(36) John is going to, *I think it's Chicago* on Sunday.

Corver & Thiersch (2001) also propose a relative clause analysis of such constructions. Regardless of the technical details of their analyses, the analyses are problematic. First, a restrictive relative clause rejects a speaker-oriented adverb (see (12a)), but a Graft construction allows such an adverb, as seen in the CC in (12b), and another Graft construction in (37) (Kluck 2011: 65).

(37) Bill stole [{*frankly/honestly*}, I thought it was just a dollar] from his grandma.

Second, an I-Clause, as a secondary illocutionary expression, may have an independent illocutionary force (Kluck 2011: 68), but a restrictive relative clause may not. In the Graft construction in (38), the I-Clause and the Host Clause are interrogative and declarative clause, respectively.

(38) Bob found – [was it a Stradivarius?] – in his attic.

I thus conclude that a Graft construction cannot be derived from a relative clause construction. See Kluck (2011: 19–25) for arguments against various versions of relative clause approaches to a WH type of Graft Constructions such as a TFR construction.

6.3 The predicate ellipsis analysis

Can the nominal in an Adv Conjunct be an argument of any deleted verbal expression? Is (39b) the structure of (39a), taking the underlined part as the antecedent of the deletion?

(39) a. John, Bill and *maybe Mary* went to the store.
b. [John, Bill ~~went to the store~~] and [maybe Mary went to the store].

Such an analysis has been proposed in Bogal-Allbritten & Weir (2017) for weak CCs. The first main problem of this analysis is seen in the scope of the adverb in a CC (Zwart 2023). In 2.1, I have argued that the adverb does not scope over the verbal expression in the Host Clause. However, *maybe* in (39b) does scope over a verbal expression. The second main problem is seen in a collective predicate. As pointed out by Collins (1988: 18), if the predicate of a clausal conjunct were deleted, the clausal coordination analysis would fail to explain the occurrence of a CC with a collective or cumulative predicate, as seen in (40a). This example cannot come from the ungrammatical (40b), where VP ellipsis occurs in the first clausal conjunct. See Vicente (2013) for more problems of the clausal coordination analysis.

(40) a. Chris, Mary and *maybe Bill* got together yesterday.
b. *Chris, Mary ~~got together~~ and *maybe Bill* got together yesterday.

The failure of the clausal coordination analysis challenges the hypothesis that nominal coordination must be derived from clausal coordination (Chomsky 1957; Gleitman 1965; Schein 2017).

The collective predicate problem is also seen in a cleft analysis of CCs. Thus, (41a) is not derived from (41b). Also, an Adv Conjunct is not focused (Chris Collins, *p.c.*).

(41) a. John, Sue and *maybe Mary* got together yesterday.
b. *John, Sue and [maybe IT BE Mary WHO ~~got together~~] got together yesterday.

Moreover, Wilder (1998: 195) claims that the Host Clause in (42a) is [*John bought a guitar*], and the predicate in the TFR is deleted, as shown in (42b) (also see (28)).

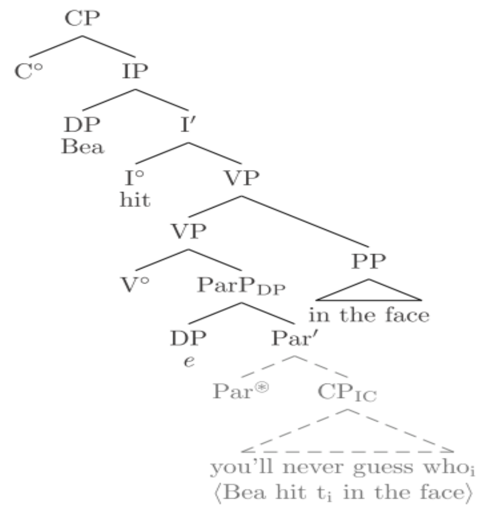
(42) a. John bought what he took to be a guitar.
b. John bought [what he took to be ~~a guitar~~] a guitar.

But (42a) does not entail [*John bought a guitar*]. I thus do not adopt this analysis.

Also, the I-Clause of Graft constructions is not a deleted copy of the Host Clause, as in (43b) for (43a) (Kluck 2011: 298). In (43b), CP_{IC} is the I-Clause, and the clause in the angled brackets is identical to the Host Clause and gets deleted, after *who* moves out.

(43) a. Bea hit *you'll never guess who* in the face.

b.



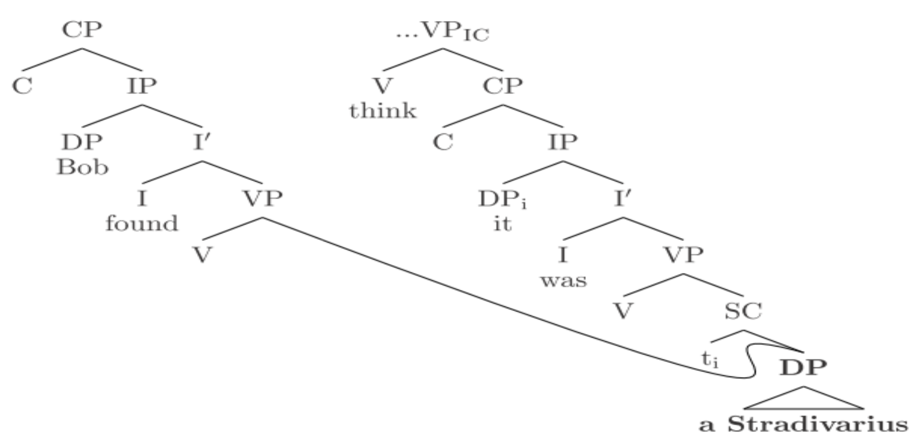
In (43b), the verb takes a DP-like phrase ($ParP_{DP}$) as its complement. In this $ParP_{DP}$, the null *e* correlates to a pronominal form such as *someone* (p. 293). This *e* and the CP_{IC} look like two conjuncts. But the relation between them does not look like that between two conjuncts (cf. Kluck 2011: 292), since a pronoun and a clause are semantically different and thus they do not conjoin. The relation is not that between a Head and its non-restrictive relative clause, either, since the latter needs a relative pronoun. Also, the CP_{IC} does not look like a modifying clause of *e*, since it does not tell us anything about *e*. The semantic relation between *e* and the CP_{IC} is not clear.

6.4 The multidominance and deletion analyses

A multidominance structure (Moltmann 1992) has been proposed for Graft constructions (Wilder 1998, de Vries 2009). For example, (44b) is the assumed structure for (44a): *a Stradivarius* is both the object of *found* and an embedded predicate in the IC (I-Clause).

(44) a. Bob found *I think it was* a Stradivarius.

b.



In (44b), the two daughters of the VP in the Host Clause are both terminal nodes, forming a constituent *found a Stradivarius*. (44b) represents two complete clauses:

(45) Bob found a Stradivarius. I think it was a Stradivarius.

But (44a) does not entail the first clause of (45) (Kluck 2011: 246). The DP *a Stradivarius* is not the object of *found* at all. The problem is similar to the one in (42b).

In addition to this entailment problem, the multidominance analysis also has a binding problem and a scope problem with respect to modal and negation (Vicente 2013). Also, since the analysis assumes that the I-Clause and the Host Clause of a CC are combined in a clause level, it shares some problems with the predicate ellipsis analysis (§6.3). Also see Kluck (2011: 32f) for her critical comments on Guimarães's (2004) multidominance analysis of Graft constructions.

Compared to the proposal in §5.3, these alternative analyses are empirically inadequate. Theoretically, in my analysis, the functional element J is recognized out of parenthetical constructions, unlike the construction-specific projections such as ParP (de Vries 2009), seen in (43b), Colon Phrase (Koster 1999), and πP (Gärtner 2001).

7. Conclusions

To answer the question how a sentential adverb is combined with a nominal in a CC, I have argued that the adverb modifies a reduced I-Clause. Also, this clause occurs in a nominal position, because the nominal in the clause moves covertly and labels an expression that contains the I-Clause. To answer the question how the weak and strong readings of a CC are computed in syntax, I have argued that a null HAVE and a null BE occur in the I-Clause, respectively. The analysis also derives other Graft constructions, including TFR constructions. It avoids the problems of the alternative syntactic analyses.

In developing a plausible derivation of CCs and other Graft constructions, this research implements the J-theory, which is established in analyzing coordination and modification constructions. Thus, an observed category mismatch is explained by the occurrence of a categoryless functional element, which makes the unexpected category invisible to a higher element.

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